

MATH 231 – Linear Algebra I

Topics Not Guaranteed to Be Covered

Fall 2026 ■ working draft — not finalized

This document is not finalized. Every topic listed here is one the course *may* reach, not one it promises to. Nothing here is part of the syllabus and none of it should be studied in expectation of assessment unless it has been officially announced. Where this document and an official course document disagree, the official document is correct.

The course progression is deliberately broader than one semester, so that topics can be selected by interest and available time rather than fixed in advance. This document collects, in one place, everything across the four arcs that falls outside the guaranteed core — so the boundary between *will cover* and *might cover* is visible at a glance instead of scattered across twelve units.

Across the arcs

These remain in the progression document, tagged *Recommended* or *Enrichment*. They are gathered here because a tag buried in a unit is easy to miss.

Topic	Unit	What it would add
Quaternions	3	Non-commutative multiplication and 3-D rotation composition; the natural sequel to complex numbers as an algebra of rotations.
LU with partial pivoting	7	Existence/uniqueness of the factorization once row interchanges are allowed; the practical form of Gaussian elimination.
Discrete Cosine Transform	10	A <i>fixed</i> orthonormal basis for compression, contrasted against the data-adaptive SVD basis; the transform JPEG actually uses.
Graph Laplacians & spectral clustering	11	$L = D - W$, and clustering a graph by the eigenvectors of its Laplacian.
Stochastic (Markov) matrices	11	Column-stochastic matrices and the dominant-eigenvector reading of a steady state.
Perron–Frobenius theorem	11	Why a nonnegative/stochastic matrix has a positive dominant eigenvector — the result that makes PageRank well posed.
Eckart–Young	11	The truncated SVD is the best low-rank approximation; the theorem underneath every compression demo.

Topic	Unit	What it would add
Randomized SVD	11	Sketching a large matrix to approximate its leading singular subspace.
Hutchinson trace estimation	11	Estimating $\text{tr}(A)$ from matrix–vector products alone.

Migrated out of Unit 12

Unit 12 (*Numerical Optimization & Machine-Learning Capstone*) previously carried the following as unit content. They have been moved here, and the unit now names them only under **If time permits**. What Unit 12 still commits to is gradient descent, Newton’s method, and Gauss–Newton/Levenberg–Marquardt for nonlinear least squares.

Topic	What it would add
Quasi-Newton BFGS	Building curvature information from successive gradients instead of forming and solving with the Hessian directly.
Support vector machines	Constrained optimization under the Vapnik objective; the maximum-margin classifier, and kernel methods reusing the similarity kernels from Unit 2. Unit 2 retains a geometric preview of the SVM margin.
Stochastic gradient descent (SGD)	Descent on noisy estimates of the gradient; the method that makes large-scale fitting tractable. Also underlies the masked-loss / ALS factorization behind the collaborative-filtering demo.
Model-selection methodology	Train/validation/test splits and cross-validation as estimates of generalization error.

Dependency worth noting. Candidate project #7 (*Mini-ML: OLS + SVM + KNN with validation*, in `MATH231_projects_and_demos`) draws on both the support vector machine and model-selection methodology. If those topics are not reached, that project is not assignable in its current form.